Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 7



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic



Extra Ordinary Individuals: Stories to Ignite Student Motivation

1. Dr. Ashok Jhunjhunwala: Background: Returned to India with a mission to bridge the digital divide, choosing social impact over a lucrative overseas career.

2. Education: B.Tech from IIT Kanpur, Ph.D. in Electrical Engineering from University of Maine (USA); later became a professor at IIT Madras.

3. Career Achievements: Led pioneering work in low-cost wireless and telecom tech, enabling rural internet access; innovated with minimal resources by mentoring students in frugal engineering.

4. Impact: Transformed rural connectivity, incubated startups, and nurtured a culture of co-creation and socially-driven innovation at IIT Madras.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

1.Dr. Anurag Kumar: Background: Joined IISc Bangalore driven by a vision to make research more collaborative and impactful across disciplines.

2. Education: B.Tech from IIT Kanpur, Ph.D. in Electrical Engineering from Cornell University, later became a senior faculty member and then Director of IISc Bangalore.

3. Career Achievements: Established interdisciplinary research centers, navigated funding challenges by building strong industry ties, and introduced open forums for academic exchange.

4. Impact: Fostered a culture of collaboration and innovation at IISc, transforming its research environment and promoting collective scientific progress.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

1.Dr. Devi Shetty: Background: Saw many poor children needing heart surgery and promised to make treatment affordable for everyone.

2. Education: Studied medicine in India and trained in heart surgery in the UK.

3. Career: Started Narayana Health, made heart surgeries cheaper by cutting extra costs and training helpers instead of hiring only expensive staff.

4. Impact: Helped thousands get heart care at low cost, reached rural areas, and built hospitals that now work with partners across the world.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

1.Dr. Sunita Narain: Background: Saw rivers getting dirty from factory waste and wanted to stop pollution.

2. Education: Studied at Delhi University and later earned a science degree in environmental studies.

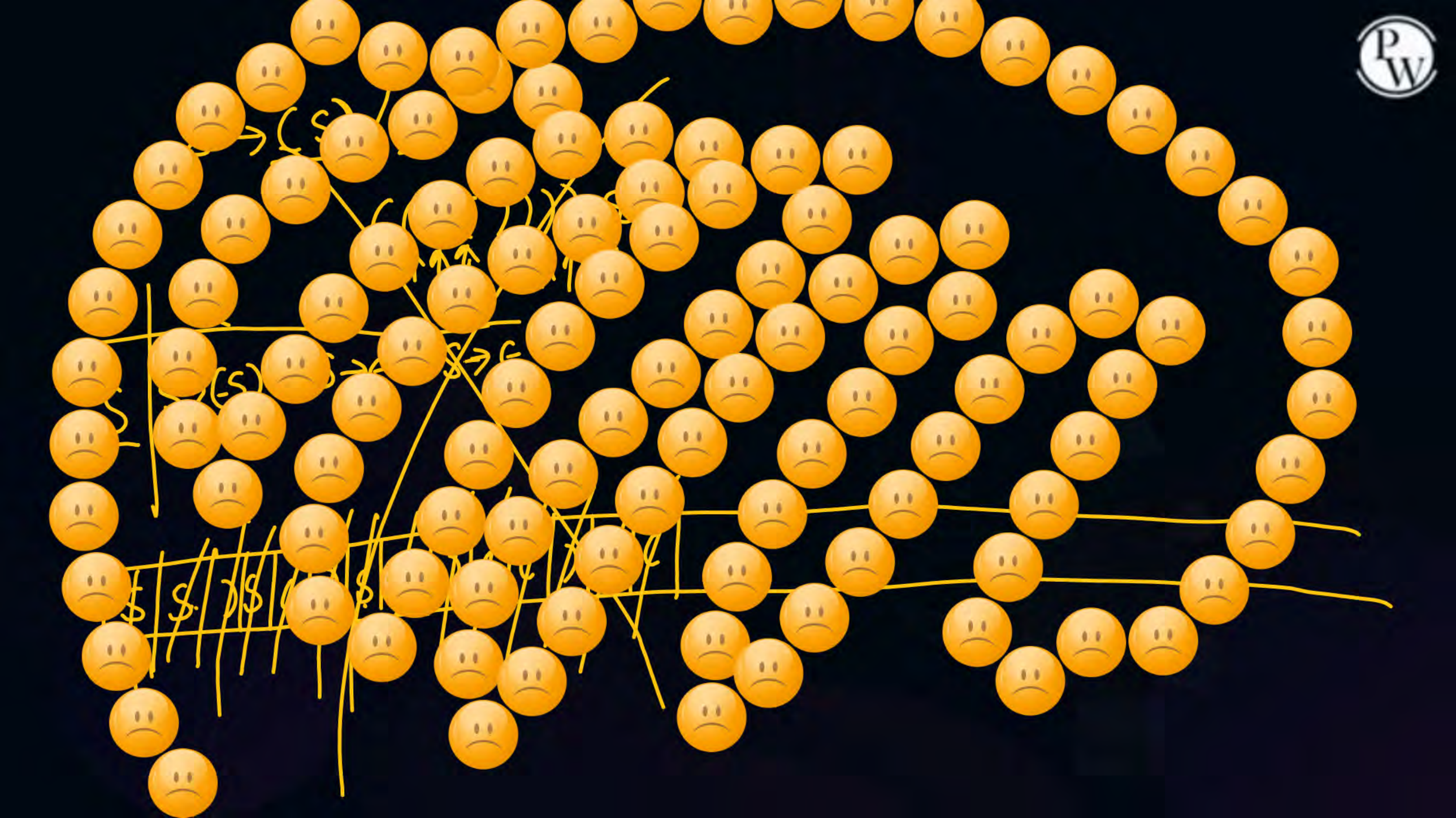
3. Career: Led environmental groups, pushed for stronger laws, and helped villagers check water quality and report problems.

4. Impact: Her reports changed court decisions and clean-up plans. She stood strong, even when big companies pushed back, to keep people and nature safe.

→ If the LL(1) parsing table doesn't contain multiple entries then such grammars are said to be LL(1) grammars

→ A grammar will definitely not be LL(1) if it is ambiguous or has left recursion or non deterministic

→ Not all grammars that are unambiguous and free from left recursion and non deterministic are LL(1) grammars



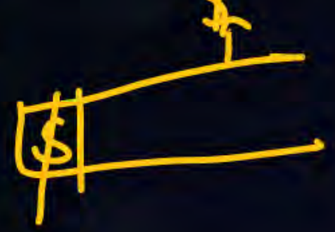
LL(1) parsing table :-

Suppose 'X' is the symbol on top of the stack and 'a' is the current i/p symbol

1) If $X == a == \$$, then successful completion
2) If $(X == a) != \$$, pop off 'X' and advance I/p pointer
3) If 'X' is a non terminal, take the ... production $m[X, a]$. Let $m[X, a]$ is $X \to UVW$, then replace X on top of the stack by UVW such that 'U' appears on top of the stack







$S \to (S) / \epsilon$

	(	)	$
S	S→(S)	S→ϵ	S→ϵ

	First	follow
$S \to (S) / \epsilon$	(, ϵ	), $

LL(1) PT will always have $ at the end of I/P.
and also $ on the bottom of the stack
Start Symbol will be pushed on to the stack

$S \to (S)$ (( )) $



$S \to (S)/\epsilon$

	(	)	$
S	S→(S)	S→ϵ	S→ϵ

(((())))$



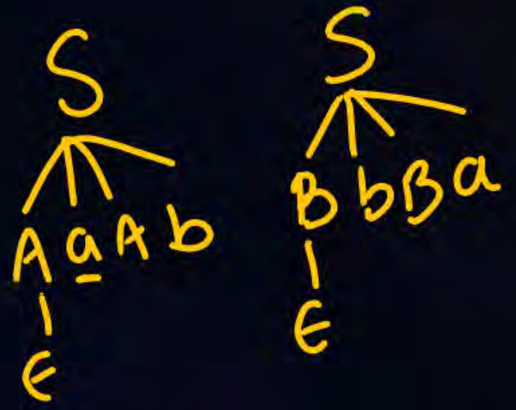
accept

	First	Follow
$S \to (S)/\epsilon$	(, ϵ	), $

LL(1) PT will always have $ at the end of I/P.
And also $ on the bottom of the stack
Start Symbol will be pushed on to the stack



	first	follow
$S \to AaAb \mid BbBa$	$\{a,b\}$	$\{\$\}$
$A \to \epsilon$	$\{\epsilon\}$	$\{a,b\}$
$B \to \epsilon$	$\{\epsilon\}$	$\{a,b\}$



LL(1) parsing table

	a	b	$
S	$S \to AaAb$	$S \to BbBa$	
A	$A \to \epsilon$	$A \to \epsilon$	
B	$B \to \epsilon$	$B \to \epsilon$	

{a} {b}

$S \rightarrow AaAb \mid BbBa$

$A \rightarrow \epsilon$ ✓

$B \rightarrow \epsilon$ ✓

LL(1)?

LL(1) ✓

How to check whether a grammar is LL(1)?

Rule 1:- A grammar without 'ϵ' productions is LL(1) if each production of the form $A \to \alpha_1 / \alpha_2 / \cdots \alpha_n$ and $First(\alpha_1)$, $First(\alpha_2)$ -- $First(\alpha_n)$ are pairwise mutually disjoint

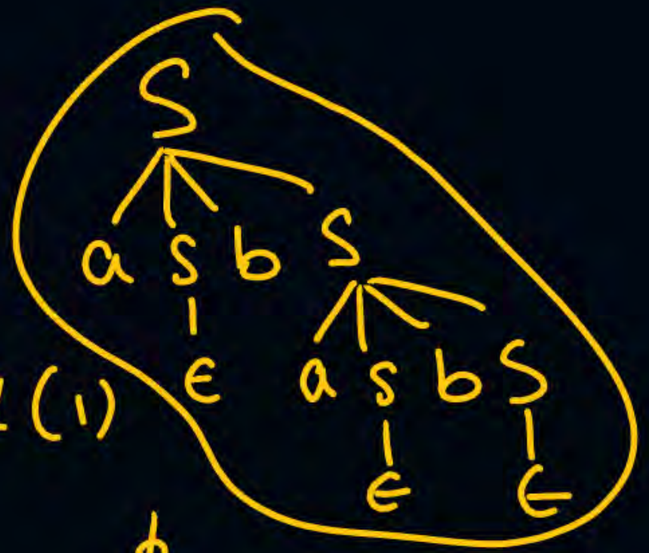
Rule:- A grammar with 'ϵ' rules is LL(1) if for each production of the form $A \to \alpha / \epsilon$, $First(\alpha)$ and $follow(A)$ must be mutually disjoint

LL(1)? (Ambiguous)

$S \to aSbS \mid bSaS \mid \epsilon$

{a} {b} {a, b, $}

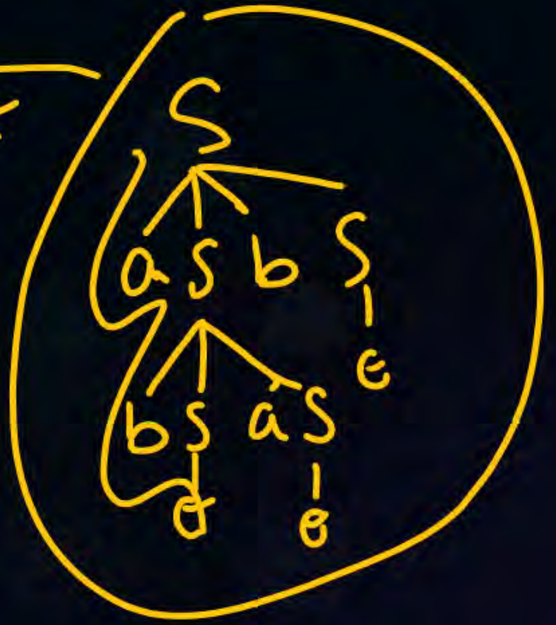
abab



✗ LL(1)

	a	b	$
S	S→aSbS / S→ϵ	S→bSaS / S→ϵ	S→ϵ

3 min



This grammar is ambiguous,

No parser can parse this grammar

	first	follow
$S \to aABb$	a	$
$A \to c / \epsilon$	(c, ϵ)	(d, b)
$B \to d / \epsilon$	(d, ϵ)	(b)

LL(1) ✓

LL(1) ?

	a	b	c	d	$
S	S → aABb				
A		A → ϵ	A → c	A → ϵ	
B		B → ϵ		B → d	

$S \to \underline{A}/a$

$A \to a$

	a	$
S	$S \to A$ / $S \to a$	
A		

S

|

a

S

|

A

|

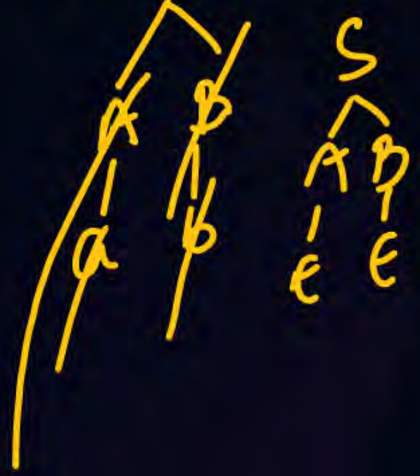
a

ambiguous

X	first	follow
S → AB	{a, b, ∈}	$
A → a/∈	{a, ∈}	{b, $}
B → b/∈	{b, ∈}	{$}

(a) $\cap(b, S) = \phi$

(b) $\cap\{\$\} = \phi$



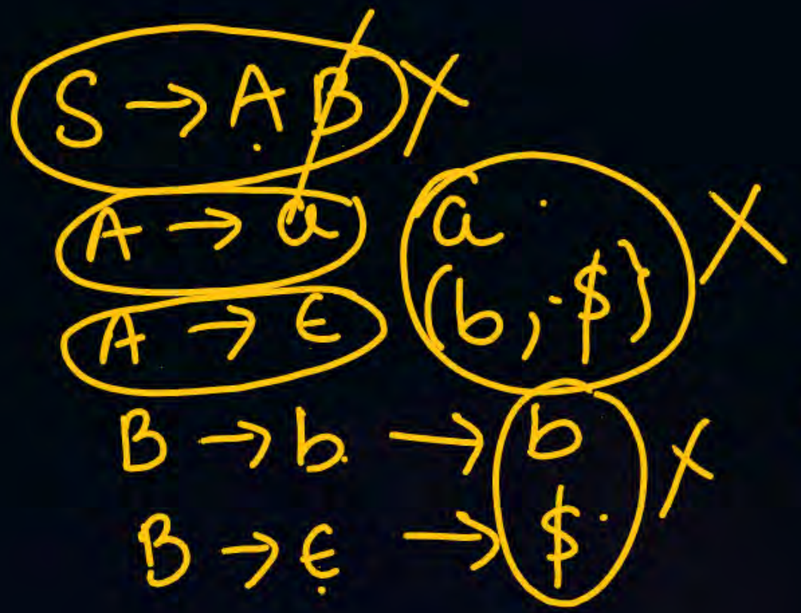
(AB) a, b, ∈

S → AB

S → AB

LL(1)

	a	b	$
S	S → AB	S → AB	S → AB
A	A → a	A → ∈	A → ∈
B	B → b		B → ∈


S → A B
A → a
A → ∈
a
{b, $}
B → b
B → ∈
b
$
X
X
X
LL(1)

	first	follow
$S \to aSA / \epsilon$	(a, ϵ)	$(\$, c)$
$A \to c / \epsilon$	(c, ϵ)	$\{\$, c\}$

$\{c\} \cap \{\$, c\}$

Not LL(1)

	a	c	$
S	$S \to aSA$	$S \to \epsilon$	$S \to \epsilon$
A		$A \to c$ / $A \to \epsilon$	$A \to \epsilon$

THANK - YOU